



Thermomechanical laser welding simulation of dissimilar steel-aluminum overlap joints

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ABSTRACT

Mixing of steel and aluminum within the weld pool during keyhole laser welding results in a complex dissimilar microstructure, which in turn, initiates a shift in weld metal mechanical properties. In this study, a numerical model for computation of distortions in laser-welded dissimilar overlap joints (austenitic stainless steel 304 – 6082-T6 aluminum alloy), which considers properties of the mixed steel-aluminum weld metal was developed. The required yield strength, Young's modulus, and strain hardening exponent of the weld metal were experimentally determined using the indentation technique coupled with energy-dispersive X-ray spectroscopy. The designed material model calculates the weld elastic-plastic properties as a function of the aluminum concentration. The softening of the alloys in the heat-affected zone was determined by physical simulations and considered as a function of maximum temperature. Computed and measured distortions showed good agreement for various welding regimes with an average deviation of 18.4%. The sensitivity analyses indicated that the application of the developed weld material model significantly improves the accuracy of the thermomechanical simulations.

1. Introduction

Multi-material design allows components weight reduction without sacrificing safety and avoiding a drastic increase in costs. By the manufacturing of such components joining of dissimilar alloys is inevitable. For a combination of steel and aluminum, the keyhole laser welding in steel-on-aluminum overlap configuration is a promising technique. Limiting weld penetration depth results in defect-free joints with high tensile strength [1–3]. However, the thermal cycle imposed on any welded parts causes non-uniform thermal expansions and contractions followed by the development of residual stresses and distortions. Since the unwanted distortions in a component are one of the most common quality deficiencies in welding production, the computational models for its prediction are of major importance for the industry.

Heat treatment during welding can highly modify microstructural state and associated mechanical properties of steels and aluminum alloys, which in turn affect the development of residual stresses and distortions. This issue obtained relatively much attention in the literature. Inherent to structural steels thermo-metallurgical solid-state phase transformation kinetics and transformation plasticity can be captured during welding simulations by a number of models [4–7]. Hamelin et al.

[8] computed residual stresses in ferritic steel weldments and demonstrated that the consideration of solid-state phase transformations leads to significant increase in prediction accuracy. In austenitic steels, the stable microstructure excludes solid-state phase transformations, and therefore, the microstructural changes do not usually require explicit consideration in computational models. For those steels, the choice of constitutive plasticity theory is of primary importance. Muránsky et al. [9] studied influence of isotropic, kinematic, and mixed isotropic-kinematic hardening models on the prediction of residual stresses in three-pass austenitic steel weld. The study showed that despite the mixed isotropic-kinematic hardening model provided the most accurate results for multi-pass welds, the isotropic hardening formalism is a suitable choice for single-pass welds. For non-ferrous materials such as 6xxx series aluminum alloys, the microstructural changes and therefore the mechanical properties are governed by precipitation kinetics, which is often described by the isokinetic dissolution model developed by Myhr and Grong [10]. Sonne et al. [11] used this softening model to consider the mechanical properties evolution while computing residual stresses after friction stir welding. The study explicitly outlined that the consideration of softening is more important for the prediction of residual stresses than the choice of the plasticity model. Bardel et al.

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[12] coupled the same metallurgical model with a thermal analysis to predict residual elastic deformations in single-pass electron beam welds. They obtained a slight transverse shift of the computed residual elastic strain profiles compared to the measured ones, which was mainly related to the oversimplification of precipitation kinetics. To get a complete distribution of precipitates for highly non-isothermal heat treatment, a more extensive KWN-type (Kampmann–Wagner Numerical) model [13] can be coupled with finite element simulations [14,15]. The so-called STAAZ model (maximum temperature austenization cooling time model) [16] considers the change in mechanical properties, induced by microstructural transformations, directly as a function of the temperature cycle, parametrized through maximum temperature, holding and cooling time. Ossenbrink [17] demonstrated the applicability of the model by performing thermomechanical welding simulations of non-alloyed low-carbon steel. The correlation between parameters of the temperature cycle and thermomechanical properties was performed by means of physical Gleeble-simulations, using typical for laser welding temperature loading. If properly calibrated, the STAAZ model is suitable for steels as well as for aluminum alloys.

Despite the thermomechanical welding simulation is an efficient tool to compute residual stresses and distortions in similar steel and aluminum weldments, it needs further adaption for dissimilar steel-aluminum (St-Al) overlap joints. The keyhole laser welding in steel-on-aluminum overlap configuration implies liquid-liquid interaction of the alloys within the weld pool and results in a complex dissimilar microstructure of the weld metal. The deficient understanding of the weld metal properties limits its consideration in thermomechanical simulations. To date, several studies have shown the association between weld mechanical properties and aluminum concentration inside the weld [1,18,19]. A linear increase in weld hardness with increasing aluminum concentration was observed. The shift in hardness was related to the aluminum concentration. In a previous work, Evdokimov et al. [20] performed laser keyhole welding of austenitic stainless steel 304 to 6082 aluminum alloy and provided first insight into the elastic-plastic behavior of the St-Al weld metal. The study showed that hardness and yield strength exhibit a significant nonlinear increase with increasing aluminum concentration, while Young's modulus and strain hardening exponent do not dependent on the aluminum concentration and remain nearly identical to the values of the steel. However, due to the limited number of investigated welds, the exact influence of the aluminum concentration on hardness and yield strength of St-Al weld metal remained unknown.

The temperature field determines the steel-aluminum mixing ratio in the weld pool as well as microstructural transformations within the steel and aluminum alloy heat affected zone (HAZ). Several works address the temperature field simulation in St-Al overlap joints. Indhu et al. [21] developed a two-dimensional multi-physical model. The authors reported that the computed weld penetration depths and ablations were in good agreement with the experiments. However, based on the provided results, the mixing ratios of steel and aluminum in the weld pool cannot be properly determined. Yan et al. [22] presented a 3D thermal conduction model. They used a conical heat source and did not consider the thermal contact in the overlap area. It was stated that the computational results were in good agreement with the experiments, however, the systematic quantitative analysis of either temperature cycles or weld shapes was not provided. In a previous work, Evdokimov et al. [23] reported that valid computation of temperature field and St-Al weld shape requires the superposition of two volumetric heat sources and the melting state-based thermal contact formulation in the overlap area. The computed and measured weld shapes were in a good agreement and allowed good prediction of St-Al dilution in the weld. However, approximation of the aluminum distribution inside the welds was not performed.

A limited number of studies on thermomechanical simulation of laser-welded dissimilar overlap joints with mixing in the weld pool is available. Hartel et al. [24] conducted the one for overlap aluminum-copper (Al-Cu) joints. The properties of dissimilar Al-Cu welds were

Table 1

Nominal chemical composition of steel in wt% [25].

Alloy	C	Si	Mn	Cr	Ni	P	S	Fe
304	0.08	≤1	≤ 2	18-20	8-10,5	≤ 0.045	≤ 0.03	Bal.

identified using a two-dimensional finite element model of the overlap joint shear test. Parameters of the defined weld material model were varied until the computed force-displacement curves matched the experimentally measured ones. The identified parameters were used for further 3D thermomechanical simulations. It was stated that introducing the Al-Cu weld material model improves the prediction of distortions in the Al-Cu overlap joints.

The purpose of this paper is to design a thermomechanical model to predict distortions in laser-welded St-Al overlap joints. The experimental work includes:

- determination of steel and aluminum softening in HAZ using physical simulations,
- characterization of the St-Al weld metal elastic-plastic properties as a function of the aluminum concentration by combining nanoindentation technique and EDS analysis.

The modeling approach includes the development of:

- a thermal finite element model that allows prediction of temperature distribution and weld shape in St-Al joints,
- material models to describe the steel and aluminum alloy mechanical properties within the HAZ during cooling,
- a weld material model that approximates the aluminum distribution inside the St-Al welds based on the predicted weld shapes and determines the properties of the St-Al weld,
- a thermomechanical finite element model that computes the distortions for the predicted thermal loading considering softening within the HAZ and the formation of the weld metal.

2. Experimental investigations

2.1. Welding trials

The keyhole laser welding of austenitic stainless steel 304 to 6082-T6 aluminum alloy in steel-on-aluminum overlap configuration was performed using a disk laser Trumpf TruDisk 16002. Tables 1 and 2 present the chemical compositions of the alloys.

Fig. 1 shows the dimensions of the joint and a scheme of the clamping. The sheets were fixed on both sides by the clamping jaws so that the optical access to the bottom surface of the sheets was available for the distortion measurements. The double-sided clamping is required to maintain the close contact of the sheets during the welding.

Welding parameters were experimentally adjusted to reach different penetration depths of the St-Al weld and hence various mixing ratios of the alloys inside the weld pool. Table 3 summarizes the welding experiments. The sheets were welded without shielding gas.

The temperature measurements were performed during the welding using Ni/CrNi type K thermocouples. The thermocouples were placed on the upper surface of the steel sheet and on the bottom surface of the aluminum sheet. The exact lateral distances to the weld centerline were measured using a light microscope after the trials. The welding distortions were determined optically, based on stochastically distributed gray value patterns on the bottom side of the sheets, using ARAMIS® (GOM, Braunschweig, Germany). The system captures the 3D coordinates of discrete points over the time. Displacements and distortions are calculated based on the difference between initial and final positions of the points. Fig. 2a presents a scheme of the optical measurements. The cameras record the displacements on the bottom surface. Fig. 2b exemplary shows the measured out-of-plane displacements (along z-axis) of

Table 2
Nominal chemical composition of aluminum alloy in wt% [10].

Alloy	Si	Mn	Fe	Cu	Mg	Zn	Ti	Cr	Al
6082-T6	0.7-1.3	0.4-1	<0.5	<0.1	0.6-1.2	<0.2	<0.1	<0.25	Bal.

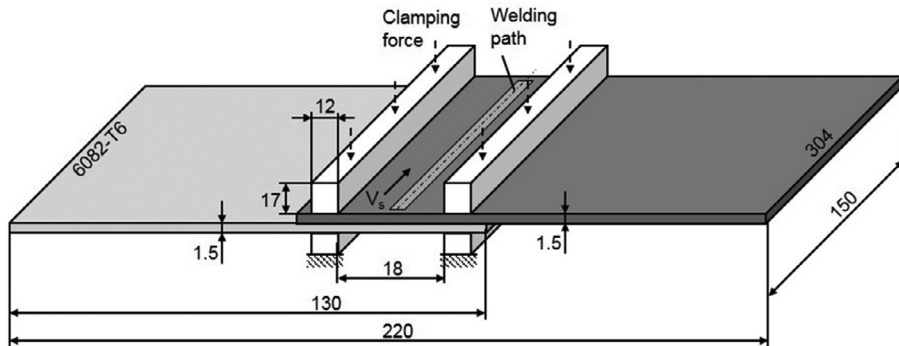


Fig. 1. Schematic illustration of welding trials.

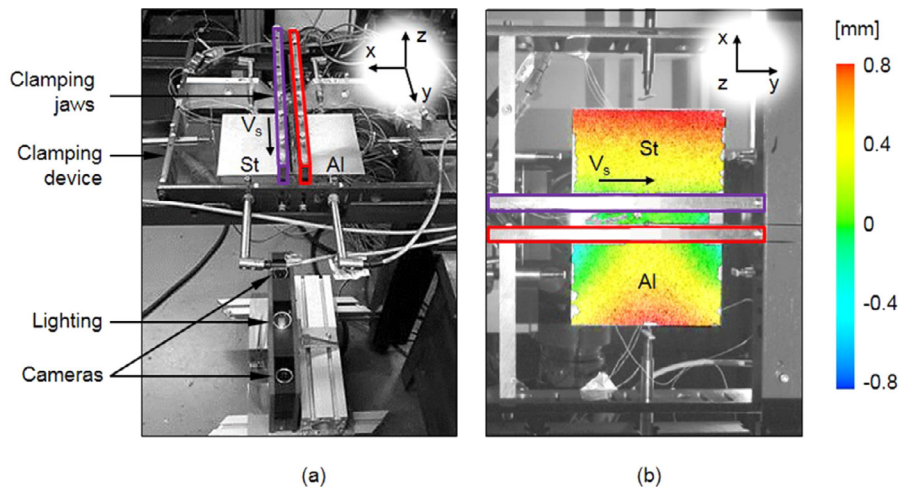


Fig. 2. Scheme of optical measurements (a). Measured residual out-of-plane displacements (b).

Table 3
Summary of welding experiments¹.

Welding regime	Welding speed, m/min	Laser power, kW
1	5.4	3.75
2	4.8	3.75
3	4.2	3.75

¹ Two trials were performed for each welding regime.

the sheets after welding and removing the clamping jaws, i.e., residual displacements.

2.2. Characterization of St-Al joint mechanical properties

Fig. 3 shows domains in the St-Al joint, which undergo microstructural evolution. The St-Al mixing in the weld pool causes a shift in the mechanical properties of the weld. The thermal impact leads to microstructural transformations and changes in associated thermomechanical properties in the steel and aluminum HAZ. The evolution of the thermomechanical properties in each part of the joint needs to be reflected in the numerical model to increase the computation accuracy.

The mechanical response of a material to heat loading can be described by its thermal expansion coefficient and elastic-plastic behavior.

The elastic-plastic behavior is defined by the basic mechanical properties such as Young's modulus, yield strength, and strain hardening exponent. In the following parts of the work all three domains are characterized.

2.2.1. HAZ_{st} and HAZ_{al}

The thermomechanical properties of the steel as well as the aluminum alloy were determined using the physical simulation center Gleeble 3500. Typical laser welding temperature cycles with various maximum temperatures were applied to large-volume flat samples to obtain divers microstructures. Since no phase transformations occur in the austenitic stainless steel due to the stable austenite microstructure, the influence of the cooling rate can be neglected. The transformation of austenite into ferrite is theoretically possible, however in practice, it is very unusual for austenitic stainless steels [25]. The test program for both, the steel and the aluminum alloy, includes determination of the heat expansion coefficient, stress-strain curves as a function of temperature during heating, and stress-strain curves as a function of temperature during cooling from different maximum temperatures (T_{max}). The samples were heated at a rate of 1000°C/s. Cooling took place through conductive heat dissipation via the clamping jaws. For the temperature measurement, a thermocouple was placed in the center of the sample, which also served as a signal transmitter to the temperature control loop.

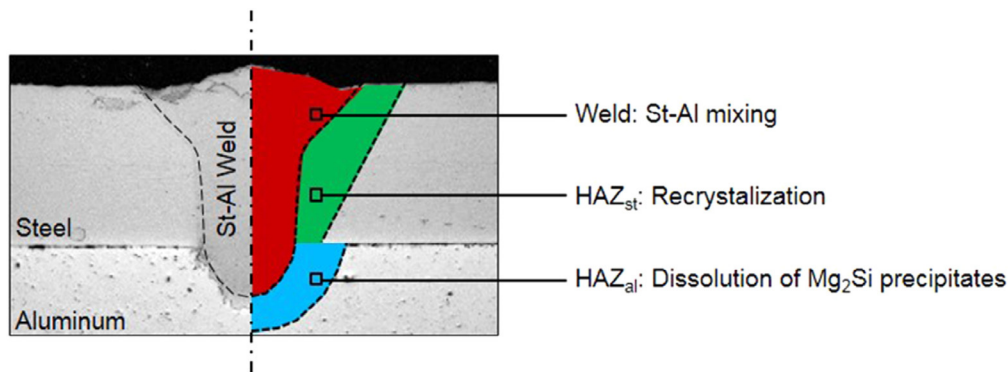


Fig. 3. Microstructural evolution in the St-Al joint.

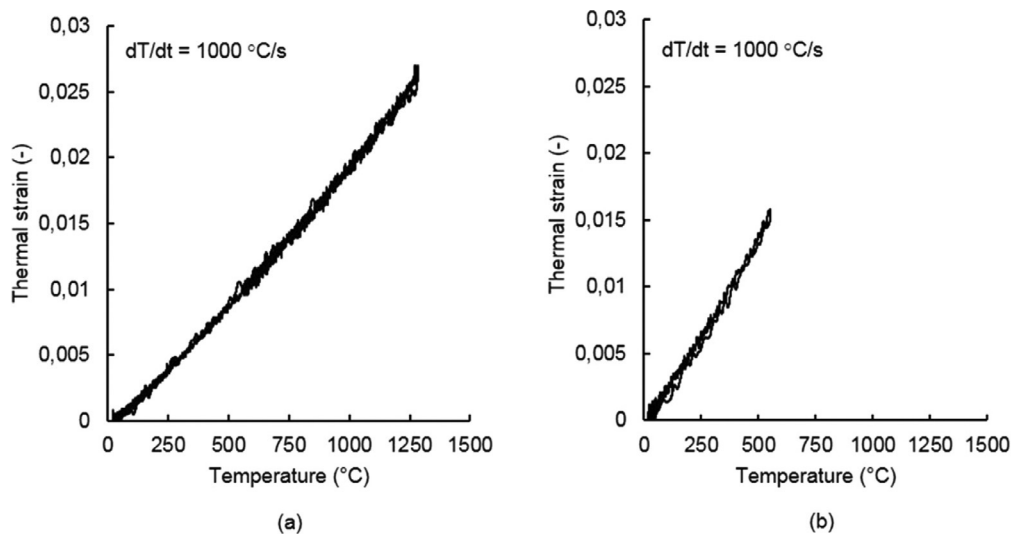


Fig. 4. Measured thermal expansion of 304 steel (a) and 6082-T6 aluminum alloy (b).

The thermal expansion was measured contactless by a laser scanner over the sample width in the test area. For the determination of the stress-strain behavior during heating, the samples were heated rapidly to the test temperature and pulled under isothermal conditions at a constant strain rate of 0.1 s^{-1} . To determine the stress-strain behavior during cooling, the samples were heated up to the required maximum temperatures, then rapidly, without holding time cooled down to the test temperature and subjected to the tensile test under isothermal conditions and constant strain rate of 0.1 s^{-1} .

Fig. 4a shows the thermal strain of the steel as a function of temperature. As expected, the austenitic steel exhibits no microstructural transformations, which affect the thermal expansion. The constant thermal expansion coefficient $\alpha_{st} = 18.3 \cdot 10^{-6} \text{ K}^{-1}$ was taken for the simulations. The summary of the yield strength values under different testing temperatures for different maximum temperatures is provided in Fig. 5a. The steel samples, heated above 700°C , exhibit a slight softening. This can be explained by the heat influence on the work-hardened microstructure, formed during the sheet manufacturing process. The dilatometer curve of the aluminum alloy also showed no deviation from linearity during heating and cooling (see Fig. 4b). The constant thermal expansion coefficient $\alpha_{al} = 26.6 \cdot 10^{-6} \text{ K}^{-1}$ was taken for the simulations. The summary of the yield strength values under different testing temperatures for different maximum temperatures is shown in Fig. 5b. The aluminum samples heated above 100°C , exhibit significant reduction in yield strength. The

softening of the age-hardened aluminum sheets is related to the transformation of the strengthening precipitates.

2.2.2. St-Al weld

The mechanical properties and corresponding chemical compositions in various small volumes within the St-Al welds were determined. Since the distribution of aluminum within the welds is not ideally homogeneous, these volumes represent St-Al weld metal with different aluminium concentrations. Indentation tests with a maximum load of 1000 mN were performed to determine the mechanical properties. The procedure based on reverse algorithms [26,27], described in our previous work [20] was used to extract strain hardening exponent and yield strength.

Outliers were excluded to avoid the artificial influence on the final function. The chemical compositions were analyzed as close as possible to the indentation imprints using scanning electron microscopy (SEM) from TESCAN (Brno, Czech Republic) with an energy dispersive X-ray (EDS) analysis facility (Oxford Analysis, Wiesbaden, Germany). Fig. 6 visualizes the proposed approach. The surface topography of the sample was obtained by KEYENCE 3D surface profiler VR-3000 (Neu-Isenburg, Germany) with a minimum resolution of 0.5 nm. Due to an inability to measure the chemical composition at the exact same location as the imprint, because of significant plastic deformation of the surface, the aluminum concentration measured close to the imprint is assumed to be the

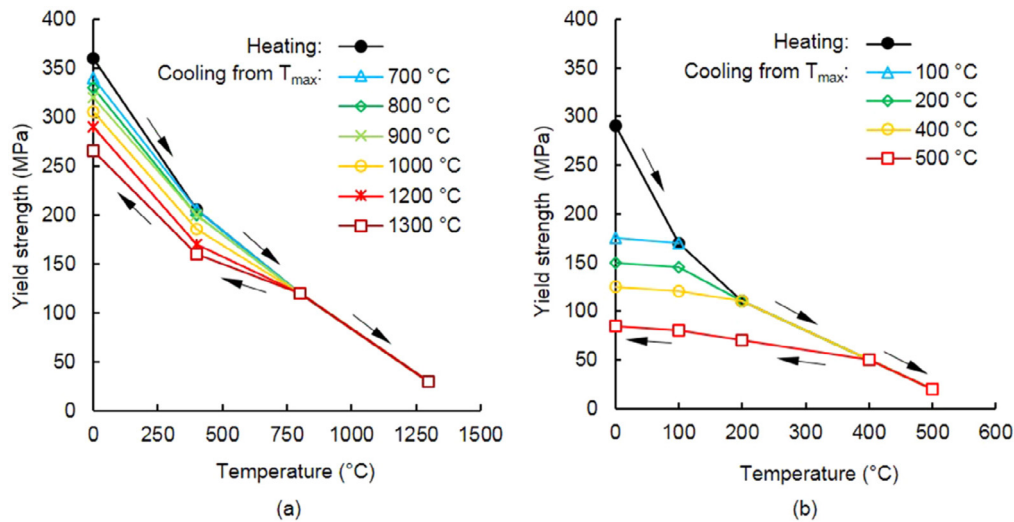


Fig. 5. Measured yield strength as a function of test and maximum temperatures of 304 steel (a) and 6082-T6 aluminum alloy (b).

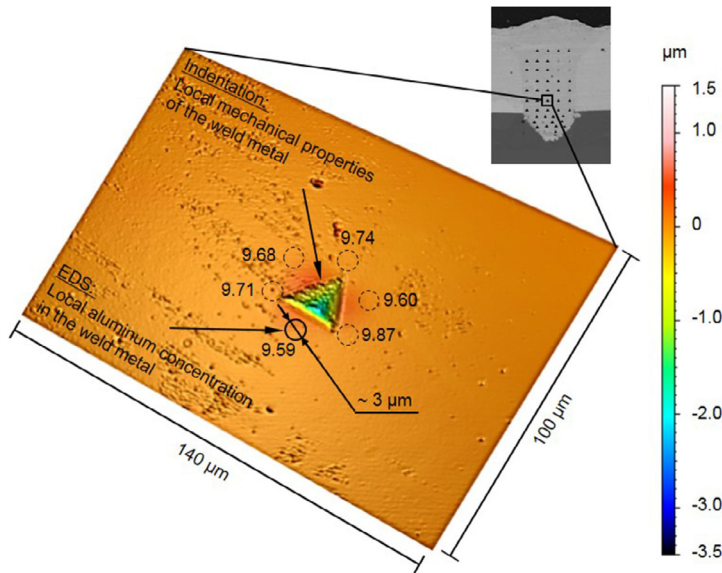


Fig. 6. The approach to determine the mechanical properties of the St-Al weld metal as a function of the aluminum concentration. Aluminum concentrations are given in at%.

concentration in the position of the imprint (black number in Fig. 6). To validate this assumption, extensive EDS measurements were performed for a couple of imprints, including the one presented in Fig. 6. The additional measurements (dashed circles) verify the almost homogenous aluminum distribution within the small volume around the imprint.

The samples for the investigation were taken out at the central positions of the welded sheets (Fig. 7). Indentation patterns were mapped in the cross-section area of the samples as shown in Fig. 7. The proposed approach allowed full characterization of the weld metal elastic-plastic properties using only these three samples. Additionally, one sample, etched using Beraha II reagent for ~ 5–10 s was subjected to EBSD

analysis to obtain the phase fraction in the weld metal, since the fcc and bcc crystal structures can be robustly distinguished by the EBSD scan. The results of the EBSD analysis are presented below in the discussion section.

Fig. 8 shows hardness, Young's modulus, strain hardening exponent and yield strength of the St-Al weld metal as a function of the aluminum concentration. A nonlinear increase in hardness is observed. Young's modulus and strain hardening exponent of the St-Al weld metal are almost insensitive to changes in the aluminum concentration. The yield strength exhibits a nonlinear increase following the behavior of the hardness.

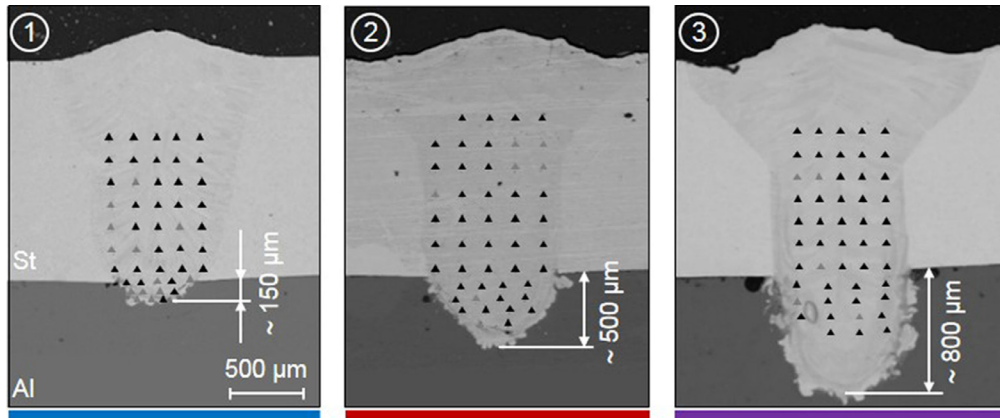


Fig. 7. Scheme of nanoindentation imprints in the investigated samples. Outliers are colored in grey. Imprints are not in scale. The samples are designated according to the corresponding welding regimes from Table 3.

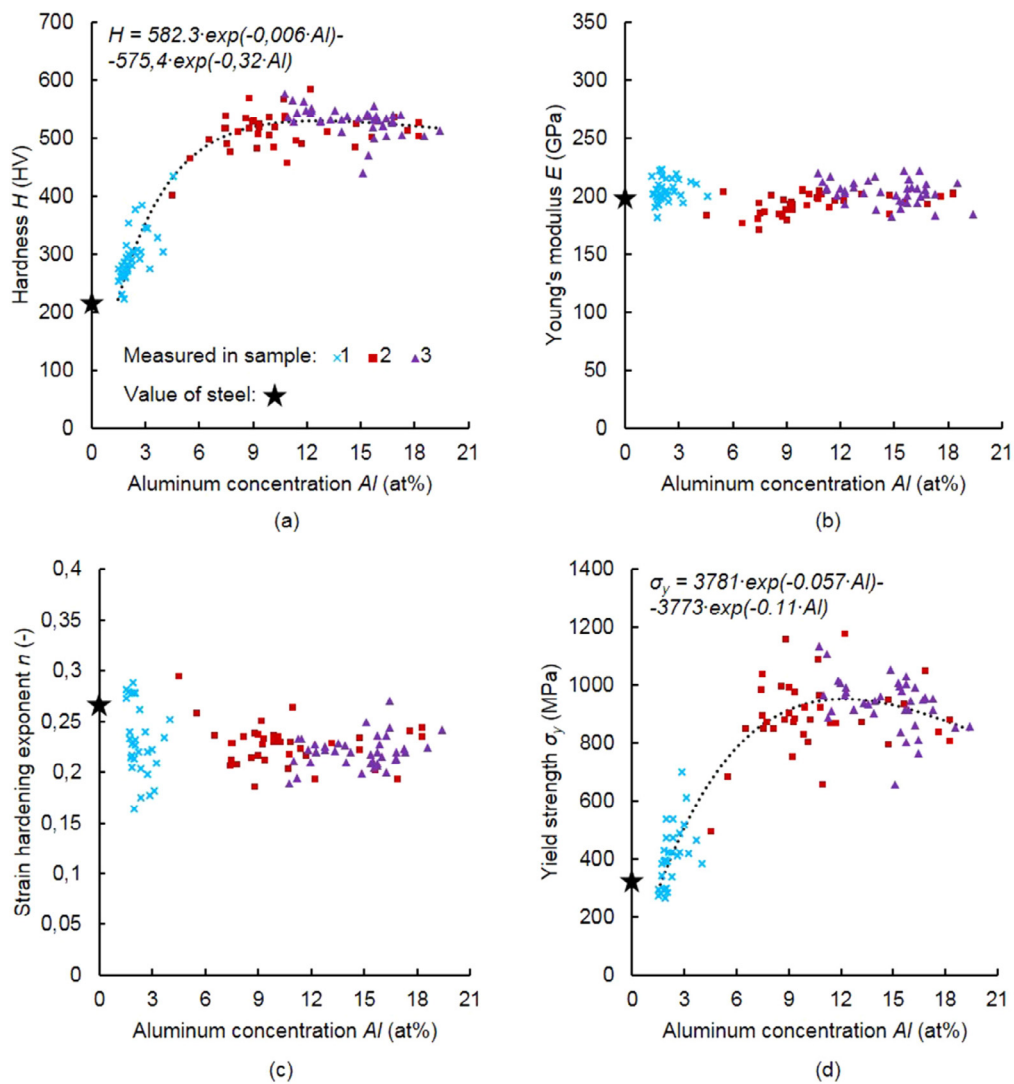


Fig. 8. Hardness (a), Young's modulus (b), strain hardening exponent (c) and yield strength (d) of the St-Al weld metal as a function of the aluminum concentration. Samples 1-3 according to Fig. 7.

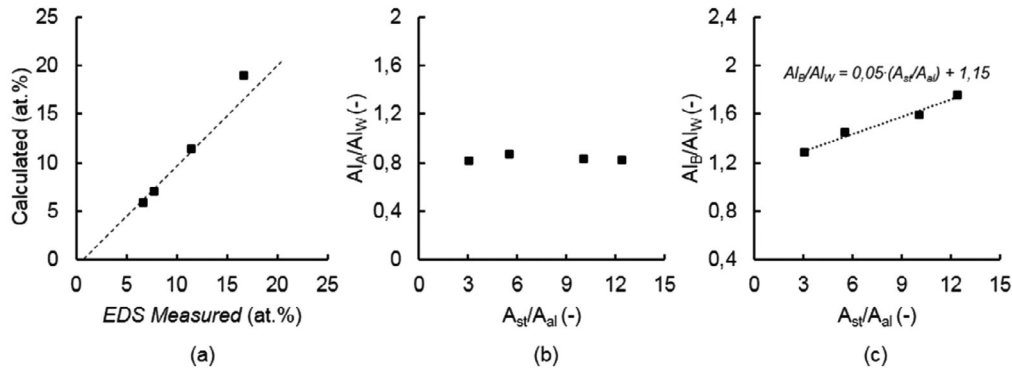


Fig. 10. Calculated versus measured average Al concentration in the St-Al welds (a). Correlation of the average aluminum concentration in the whole St-Al weld to the average aluminum concentrations in zone A (b) and zone B (c).

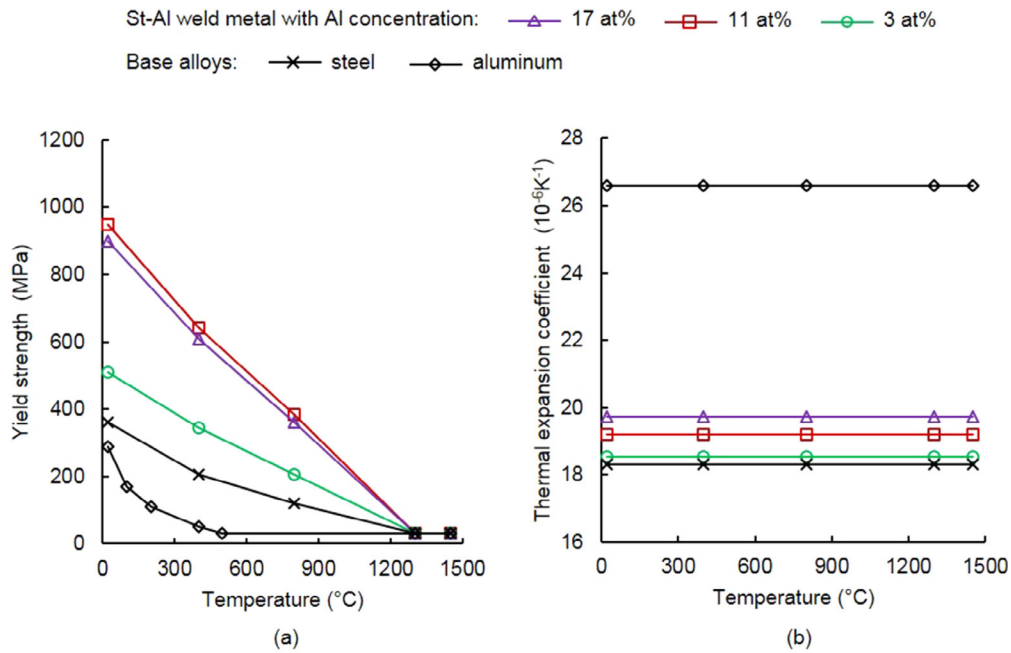


Fig. 11. Yield strength (a) and thermal expansion coefficient (b) of the St-Al weld metal as a function of aluminum concentration and temperature.

the yield strength and thermal expansion coefficient of the St-Al weld metal as functions of the temperature exemplary for three arbitrary chosen aluminum concentrations. To approximate the near liquid material state at high temperatures ($T > 1300^\circ\text{C}$), the yield strength at the point of $T = 1300^\circ\text{C}$ was artificially dropped down to 30 MPa.

The properties of the weld metal defined for thermomechanical simulations are summarized as follows:

- Yield strength, $\sigma_y - f(\text{Al}, T)$
- Young's modulus, $E - f(T)$
- Tangent modulus, $E_t - f(T)$
- Thermal expansion coefficient, $\alpha - f(\text{Al})$
- Poisson's ratio, $\nu - \text{const}$

Subsequent to the temperature simulation, the weld material model calculates the aluminum concentrations in zones A and B (see Fig. 9b)

and defines set of corresponding thermomechanical properties. While heating, the structural behavior is governed by the properties of steel and aluminum, whereas during cooling the material response within the fusion zone is switched to the weld material model.

4. Numerical model

4.1. Temperature field model

The 3D thermal conduction model based on the finite element analysis was established. The asymmetric configuration of the St-Al overlap joint requires consideration of the complete geometry of the sheets. Fig. 12 shows meshing strategy and boundary conditions of the temperature finite element model. The implementation was done exemplary using the ANSYS FE-code. The model consists of 246 338 nodes and 304

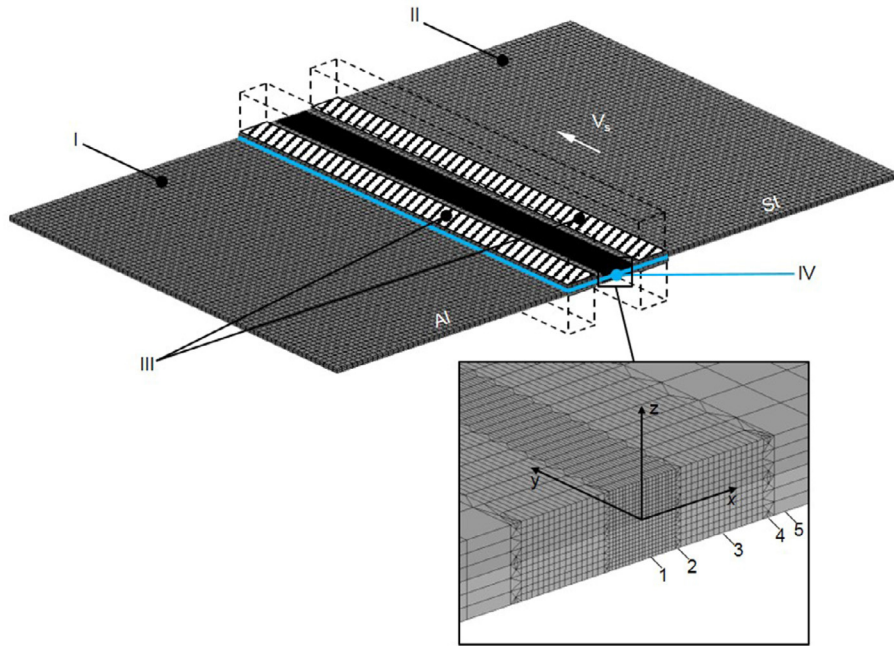


Fig. 12. Meshing strategy and boundary conditions. I, II – heat exchange with the ambient, III – heat losses into the clamping jaws, and IV – thermal contact between the sheets.

059 elements. The element size lengths were defined as $0.15 \times 0.5 \times 0.15$ mm in zone 1, $0.25 \times 1 \times 0.25$ mm in zone 3 and $2 \times 2 \times 0.5$ mm in zone 5. The transition areas 2 and 4 consist of pyramidal elements. Mesh convergence analysis was performed to ensure mesh-independent computational results. The change in physical properties of the weld metal, due to the mixing inside the weld pool, was not considered for the temperature simulation.

Heat losses were considered through convection and radiation with distinct film coefficients for steel (I), aluminum (II) and the areas where the clamping jaws are located (III). The thermal contact coefficient between the sheets in the overlap area prior to joining h_s was calculated according to [29] as described in [23]. After the sheets have been welded, the ideal thermal contact across the interface is established. In the current work, the irreversible melting state-based thermal contact was realized through the superposition of contact elements. Two contact surfaces were defined in the overlap area between the sheets. At the beginning of the computation, the contact elements with thermal conductance h_s are defined. During the computation, these elements are replaced by the elements with the ideal thermal conductance if the temperature in the contact exceeds the solidus temperature of the aluminum alloy. The superposition of two volumetric heat sources [23] was used to capture the finger-shaped weld geometry.

4.2. Thermomechanical model

For computation of distortions, a nonlinear FE model was coupled with the temperature field simulation. The boundary conditions of the thermomechanical model during welding are shown schematically in Fig. 13. On the upper surface of the joint, the normal force, induced by the upper clamping jaw, is represented by the force F_n acting at all nodes, which are under the projected area of the jaw. The value of the force F_n is calculated as the total clamping force divided by the number of nodes. On the bottom surface of the sheet, the jaws act as support preventing the movement in $-z$ direction. The influence of the friction forces between the clamping jaws and the sheets on the computation of distortions was found to be not significant and therefore neglected in the final model. It substantially speeds up the thermomechanical computations and results in better convergence of the numerical solution. In order to prevent penetration of the mesh in the overlap area, a melting state-based mechanical contact was defined. The contact surface re-

mains closed after the heating to represent the bonding of the sheets. The bonding temperature was set to the melting temperature of the aluminum alloy. Fig. 14 shows the boundary conditions after welding. The multistep approach was applied to simulate the unclamping. The force F_n is decreased at some increment for several steps until it is eliminated.

5. Results

5.1. Temperature field, weld shape and dilution

The developed FE model allows computation of full temperature history during deep penetration laser welding of St-Al overlap joints. Fig. 15 presents a plot of computed temperature cycles versus measured ones exemplary for the welding regime 2 (Table 3). The measured temperature cycles and distances from the weld centerline to the thermocouples are the averages of the two welding trials. As can be seen from Fig. 15, the computed maximum temperatures, heating and cooling rates are in good agreement with the measurements.

The plot of maximum temperatures in the cross-section allows comparison of the real and computed weld shapes. The light gray color corresponds to the area where the computed maximum temperatures exceed the melting temperatures of the alloys. The weld shapes are in a good agreement. Fig. 16 summarizes the computed maximum temperatures versus experimentally measured ones for all regimes. In general, the computations are in a good agreement with the measurements. The average deviation among all comparisons is about 4.1 %.

The dilution of the alloys is expressed according to Eq. (6) based on the ratio of the areas A_{st} and A_{al} (see Fig. 15).

$$D = \frac{A_{al}}{A_{al} + A_{st}} \times 100 \quad (6)$$

Computed versus measured dilutions are presented in Fig. 17. The post-processing routine calculates the dilutions through the number of elements belonging to A_{st} and A_{al} areas (Fig. 15). A good agreement for all regimes was observed. Special attention must be paid on the first welding regime (Table 3). Under these welding conditions, the mixing does not occur along the full length. Two out of three cross sections indicated the solid-liquid bonding without mixing between the alloys.

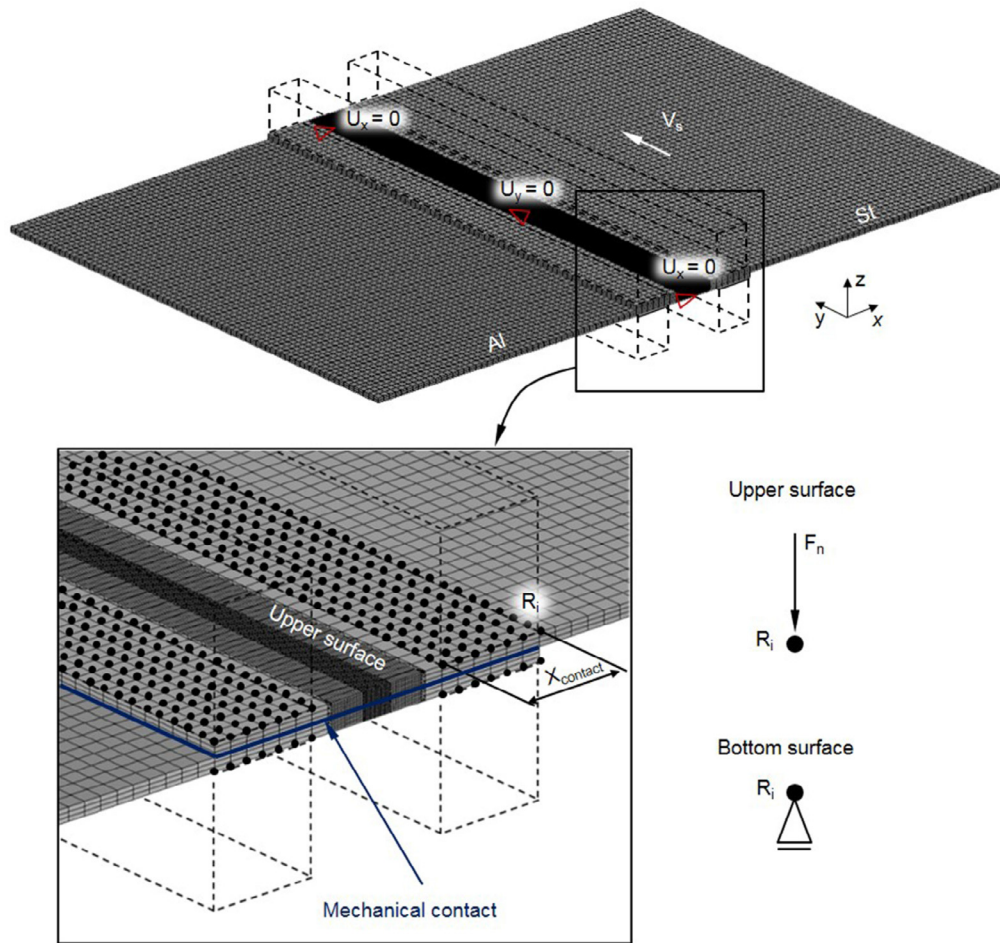


Fig. 13. Scheme of mechanical constraints during welding.

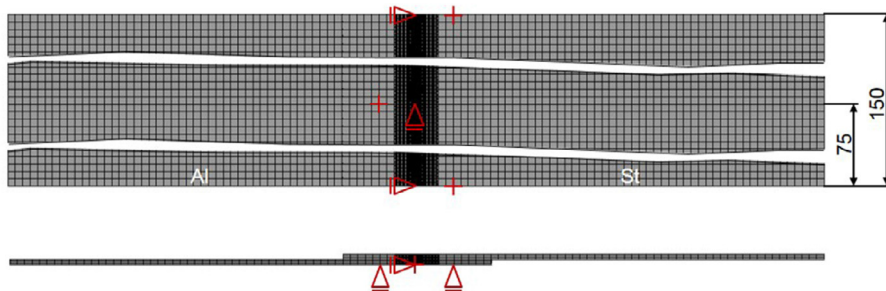


Fig. 14. Scheme of mechanical constraints after welding (unclamped condition).

5.2. Distortions

Transversal shrinkage X , longitudinal shrinkage Y_{st} , Y_{al} , and angular distortion in the central plane Z_{st} , Z_{al} were evaluated for validation of the model (Fig. 18). Fig. 19 shows measured versus computed distortions for all three regimes (Table 3).

Simulations and experiments show qualitative and quantitative agreement. The average deviation among all comparisons is 18.4 %. The

developed FE model is valid and provides relatively accurate estimation of the distortions for St-Al overlap joints. The decrease in welding speed from 5.4 m/min down to 4.8 m/min qualitatively changes the values of angular distortions (see Fig. 19). Since the shift in energy input is not essential, the difference is induced by the appearance of the St-Al weld metal. The phenomenon was well captured by the developed simulation model.

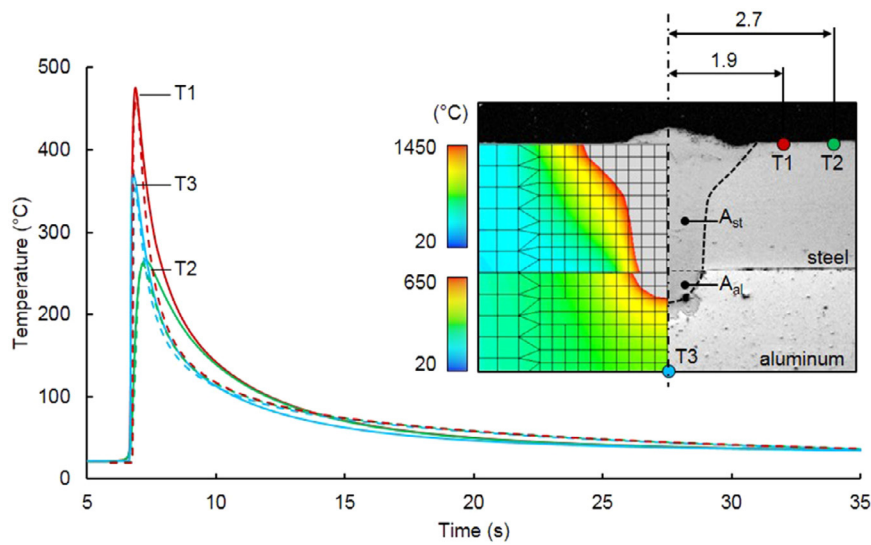


Fig. 15. Computed versus measured temperature cycles. Exemplary for regime 2 (Table 3). Solid lines – measurement, dashed lines – FEM. Distances to thermocouples are in mm.

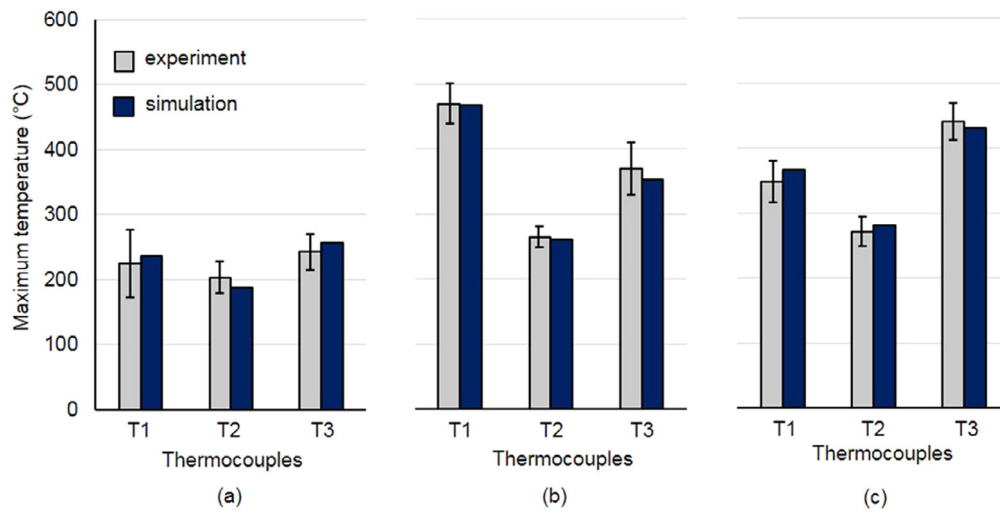


Fig. 16. Computed versus measured maximum temperatures. $P_L = 3.75$ kW.

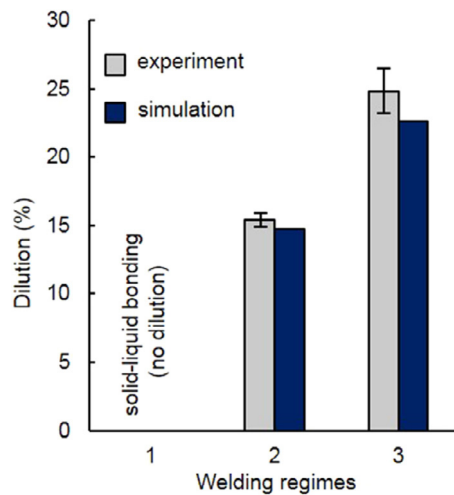


Fig. 17. Computed versus measured dilutions.

6. Discussion

6.1. St-Al weld metal

Fig. 20 compares weld metal hardness and yield strength obtained in the context of the current research versus the ones reported in the literature. Due to the lack of studies on the mixed St-Al welds, the as-cast alloys with similar chemical compositions are involved in the discussion to validate the results of the current work and to assist in understanding of the underlying strengthening mechanisms. The properties of as-cast alloys FeAl_x [30], $\text{Fe}_{40}\text{Mn}_{25}\text{Cr}_{20}\text{Ni}_{15}\text{Al}_x$ [31], $(\text{FeCoCrNi})_{100}\text{Al}_x$ [32], $(\text{FeCoMnCrNi})_{100}\text{Al}_x$ [33], and annealed alloy $\text{Fe}_{71}\text{Cr}_{10}\text{Ni}_7\text{Al}_{12}$ [34] are integrated in Fig. 20. The consistency in microstructure and mechanical properties of cast alloys and laser fabricated alloys was recently demonstrated by Li et al. [35].

The mixing of non-alloyed steel and aluminum alloy in the weld pool results in linear hardness increase of the weld metal with increasing Al concentration (Fig. 20a). It is consistent with the linear increase in yield strength of the $\text{Fe}_{100}\text{Al}_x$ alloy (Fig. 20b). It is supposed that strengthening of the simple Fe-Al system with increasing Al concentration up to 20 at% is associated with solid solution hardening [18,30]. However, the

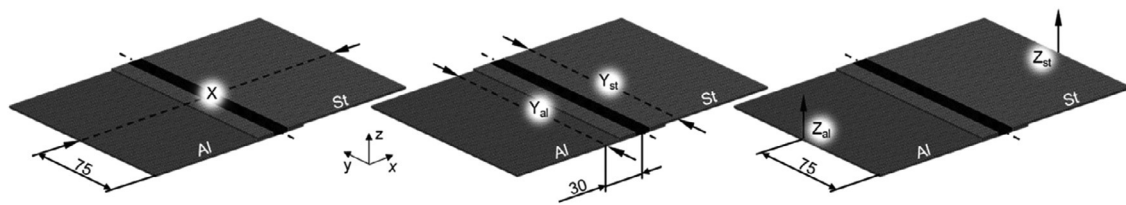


Fig. 18. Scheme of the evaluated distortions.

increase in weld hardness is significantly higher and deviates from linearity if mixing of austenitic stainless steel and aluminum alloy occurs in the weld pool. The results of the current study, as well as the ones reported by Yang et al. [36], indicated sharp hardness increase with increasing Al concentration up to a certain concentration with no further strengthening when this concentration was exceeded. High hardness of the weld metal, 400–600 HV, was also obtained after welding of stainless steel to aluminum alloy in [37,38], however, the correlation to Al concentration was not provided. This significant and nonlinear strengthening does not seem to be caused only by the solid solution hardening.

The austenitic stainless steel 304 cannot be substantially hardened by either precipitation or phase-transformation [25,39]. Therefore, the full strengthening of the St-Al weld metal is related to the addition of Al. It is well known that aluminum acts as bcc-stabilizing element and can promote precipitation reactions in stainless steels [25]. In this section the weld metal is considered mainly as a cast alloy $\text{Fe}_{70}\text{Cr}_{19}\text{Ni}_8\text{Mn}_2\text{Al}_x$, obtained by the addition of Al into the austenitic stainless steel 304. The chemical composition changes alongside with the penetration depth of the weld, as the dilution increases. Measured chemical composition of the weld metal as a function of Al concentration is shown in Fig. 21a. The data are from the EDS measurements described in Section 2.2. According to its configurational entropy, ΔS_{mix} , [40] and elements fractions, the St-Al weld metal belongs mainly to the group of non-equiatomic medium-entropy alloys (Fig. 21b) with Fe as matrix element and Cr, Ni, Mn, Al, as equiatomic alloying elements.

Zhou et al. [41] provided an extensive research on as-cast medium-entropy $\text{Fe}_{55}(\text{CrNiAl})_{45}$ and $\text{Fe}_{65}(\text{CrNiAl})_{35}$ alloys, which chemical compositions are quite similar to the considered St-Al weld metal. Both alloys consisted of Fe-Cr-rich bcc-disordered matrix with embedded coherent Ni-Al-rich B2-ordered nanoprecipitates. Ni-Al-rich B2 phase is formed since Ni and Al elements have the most negative mixing enthalpy as well as the largest difference in Pauling electronegativity [41]. The coherent precipitates, which are responsible for the high strength, arise due to a small lattice parameter mismatch between the B2 NiAl precipitates and the bcc matrix [42,43]. The similar bcc/B2 structure was also found in non-equiatomic $\text{Fe}_{36}\text{Mn}_{21}\text{Cr}_{18}\text{Ni}_{14}\text{Al}_{10}$ and equiatomic $(\text{FeCoCrNi})_{78}\text{Al}_{22}$, $(\text{FeCoCrNi})_{83}\text{Al}_{17}$ high-entropy alloys [31,35,44] as well as in a number of high-strength aged Fe-Al-Cr-Ni ferritic alloys [34,42,43,45–48].

One cross section of the St-Al weld with a penetration depth of about 500 μm was subjected to EBSD analysis (Fig. 22). The chemical composition of the tested weld metal is $\text{Fe}_{63}\text{Cr}_{17}\text{Ni}_7\text{Mn}_2\text{Al}_{11}$. The bcc structure was identified in excellent agreement with the mentioned cast alloys. The SEM analysis did not allow to detect the B2 NiAl precipitates. The size of precipitates in cast $\text{Fe}_{65}(\text{CrNiAl})_{35}$ and $\text{Fe}_{36}\text{Mn}_{21}\text{Cr}_{18}\text{Ni}_{14}\text{Al}_{10}$ alloys were found to be 100 and 160 nm respectively [31,41]. Therefore, TEM analysis is required to characterize the morphology of the precipitates within the weld metal, which can slightly differ due to the highly non-equilibrium conditions during laser welding. Nevertheless, based on the experimentally identified bcc structure as well as the similarity in chemical composition and mechanical properties with the extensively studied cast alloys, it can be suggested that the weld metal, with Al concentration of 11 at% consists of Fe-Cr-rich bcc matrix strengthened by coherent B2 Ni-Al-rich nanoprecipitates.

Microstructures and strengthening of as-cast high entropy alloys [31–33] were correlated over the wide range of Al concentrations (Fig. 20). It was confirmed that the addition of Al changes the crystalline structure from the initial fcc to the duplex fcc-bcc and subsequently to the bcc. The perfect correlation between crystalline structure and strength becomes obvious (Fig. 20). The alloys with fcc structure exhibit no significant strengthening at any Al concentration. As soon as the fcc structure is destabilized, the increase in hardness and yield strength becomes proportional to the fraction of bcc phase and can be approximately described by the rule-of-mixture [33]. The highest strength is achieved when the alloy reaches the fully bcc structure. Once the transformation into bcc is completed, further increase in Al concentration does not affect strength significantly. The same evolution of the crystalline structure with increasing Al content was also observed for laser fabricated high-entropy FeCoCrNiAl_x alloys [49,50]. It worth noting that the yield strength of $\text{Fe}_{40}\text{Mn}_{25}\text{Cr}_{20}\text{Ni}_{15}\text{Al}_x$ alloy significantly increases even at low Al concentrations (Fig. 20b). It is explained by the initial crystal structure of the alloy, which is already prone to the bcc phase formation. Therefore, even small addition of Al causes an increase in bcc fraction and, as a result, an increase in strength. A similar trend was also found outside of the high entropy alloys world. Rahimi et al. [51] demonstrated the microstructural changes caused by the addition of Al into as-cast austenitic $\text{Fe}_{67}\text{Cr}_{17}\text{Mn}_6\text{Co}_{0.45}\text{Ni}_9\text{Al}_x$ (in wt.%) stainless steels with initial hardness of 175 HV₁₀. Addition of 4 wt.% (~ 8 at. %) retained the fcc structure and did not change the hardness. Further increase up to 7 wt.% (14 at%) resulted in fully bcc matrix with a high hardness of 424 HV₁₀.

Based on the similarity with the cast alloys, the explanation of the microstructural changes in the St-Al weld metal can be suggested. The addition of Al in 304 stainless steel in the weld pool causes the transformation of the initial fcc structure into bcc and the formation of B2 Ni-Al nanoprecipitates. Since these precipitates are coherent with the bcc phase, the increase in bcc fraction results in substantial strengthening of the weld metal. Significant hardness increase at small Al concentrations indicates that even a small portion of Al is sufficient to initiate the bcc phase formation. Thus, it is supposed that the St-Al weld metal with Al concentration up to 9 at% exhibits a mixture of fcc and bcc phases with hardness and yield strength increase proportional to the fraction of the bcc phase. The strength saturation at Al concentrations beyond 9 at% is related to the formation of the fully bcc matrix.

Thus, the prediction of the St-Al weld metal yield strength requires the determination of its crystalline structure and the calculation of the precipitation strengthening. The crystalline structure of stainless steels can be evaluated using the WRC 1992 diagram [25]. However, the applicability of this diagram for medium-entropy alloys as well as for the laser welding is disputable. Li and Flores [52] showed that high cooling rates, greater than 553 K/s, can cause the formation of a bcc structure at lower Al concentrations. For high entropy alloys, the stability of fcc and bcc phases can be estimated based on the valence electron concentration VEC. Guo et al. [53] found that the fcc phase is stable at $\text{VEC} \geq 8$ and the bcc phase is stable at $\text{VEC} \leq 6.87$. A microstructure containing both fcc and bcc phases is formed if VEC is between 6.87 and 8. The VEC value of the weld metal with Al concentration of 11 at% ($\text{Fe}_{63}\text{Cr}_{17}\text{Ni}_7\text{Mn}_2\text{Al}_{11}$) is 7.27. This value is within the fcc-bcc region, however, according to the EBSD analysis, the matrix consists of the single bcc phase. The alloys

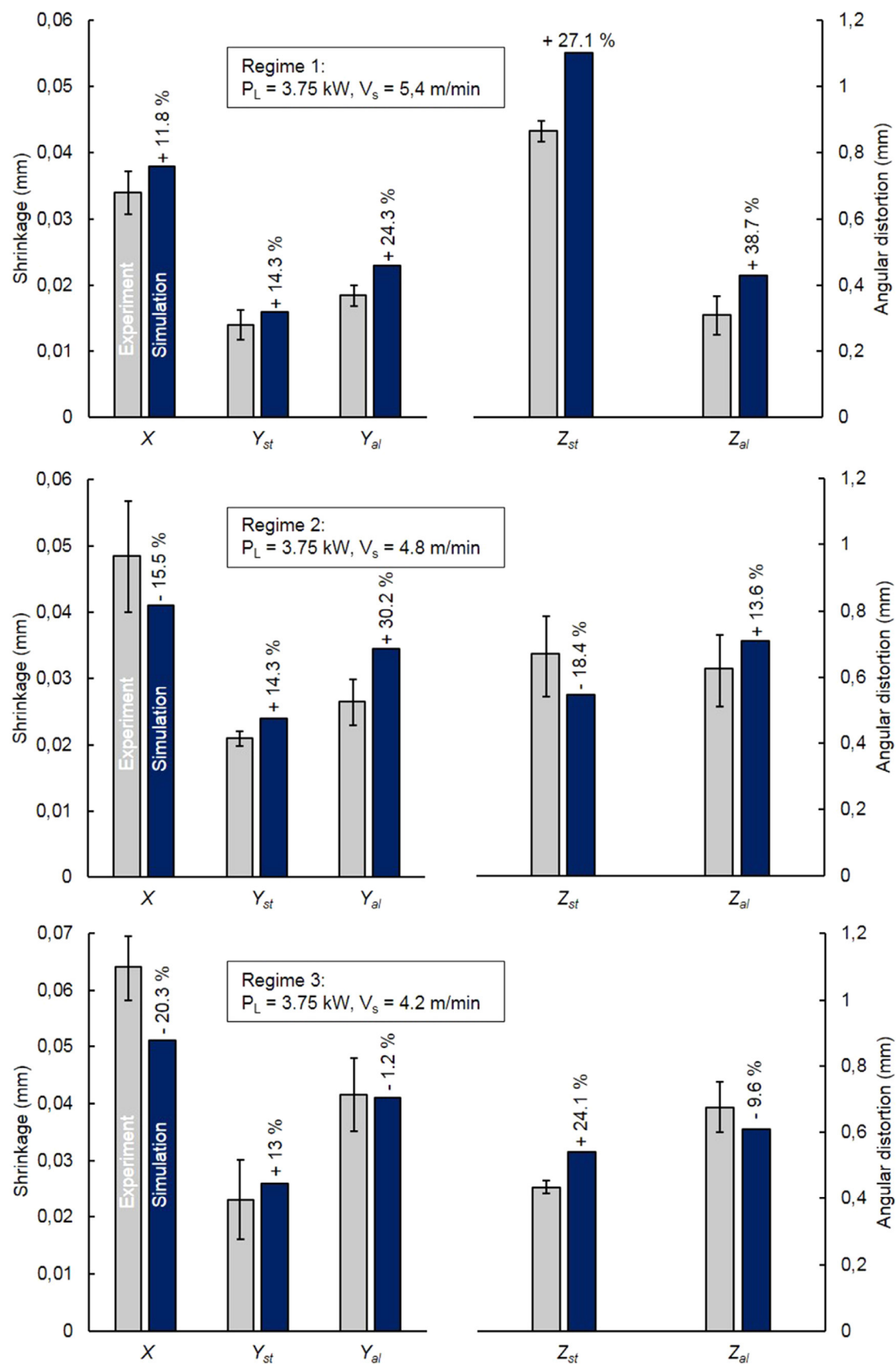


Fig. 19. Measured versus computed distortions.

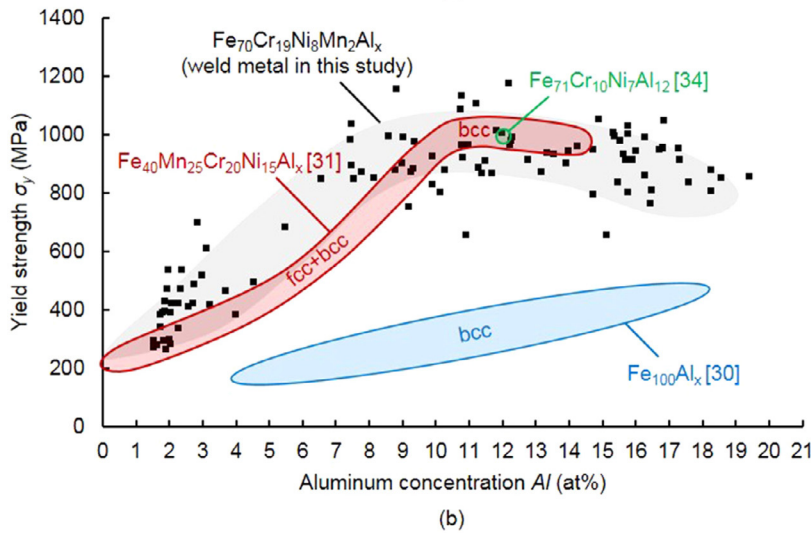
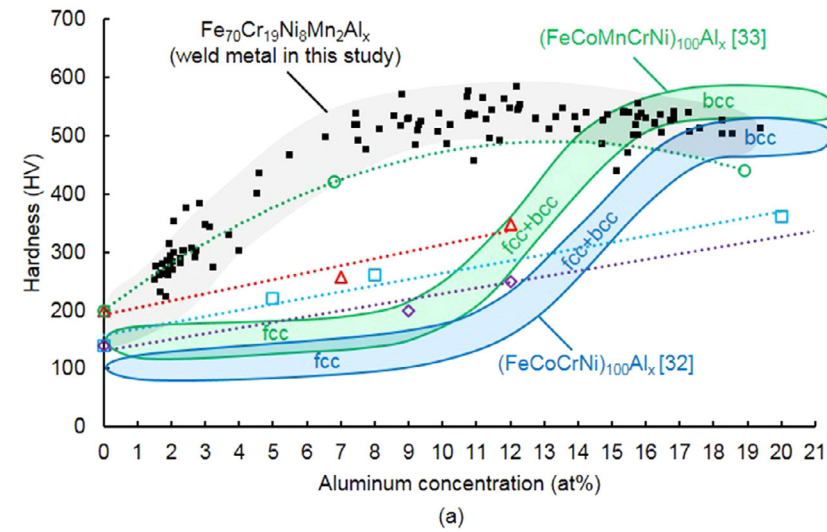


Fig. 20. Hardness (a) and yield strength (b) versus Al concentration. Weld metal arose after mixing of aluminum with non-alloyed steel ($\square$ [18], $\diamond$ [19], $\triangle$ [1]) and stainless steel ($\blacksquare$ [current study], $\circ$ [36]).

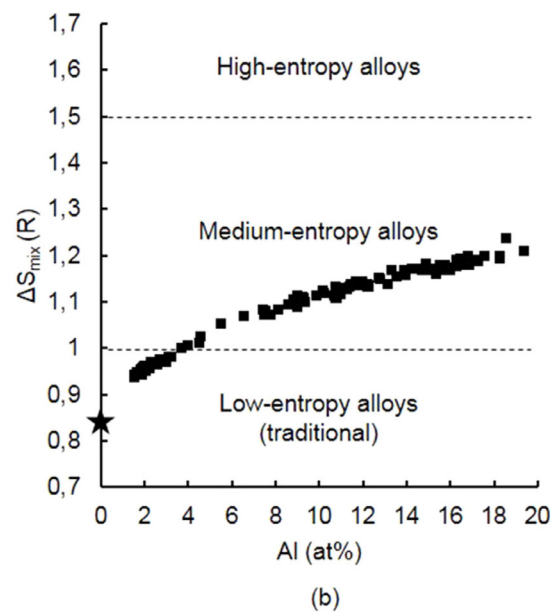
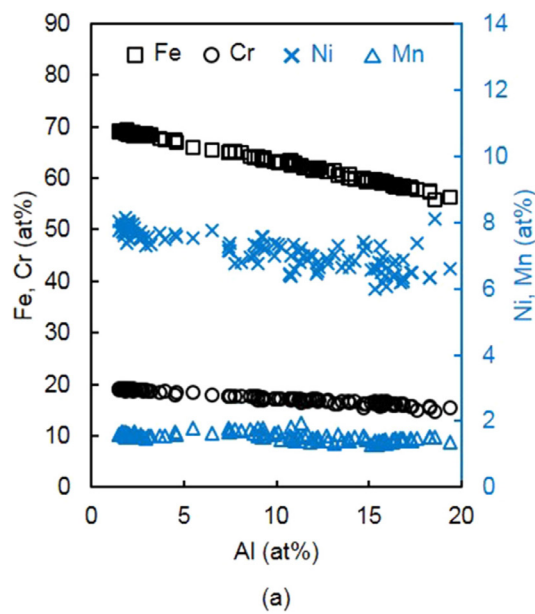


Fig. 21. Measured chemical composition (a) and calculated corresponding configurational entropy (b) of the weld metal as a function of Al concentration. Configurational entropy of stainless steel 304 ($\star$).

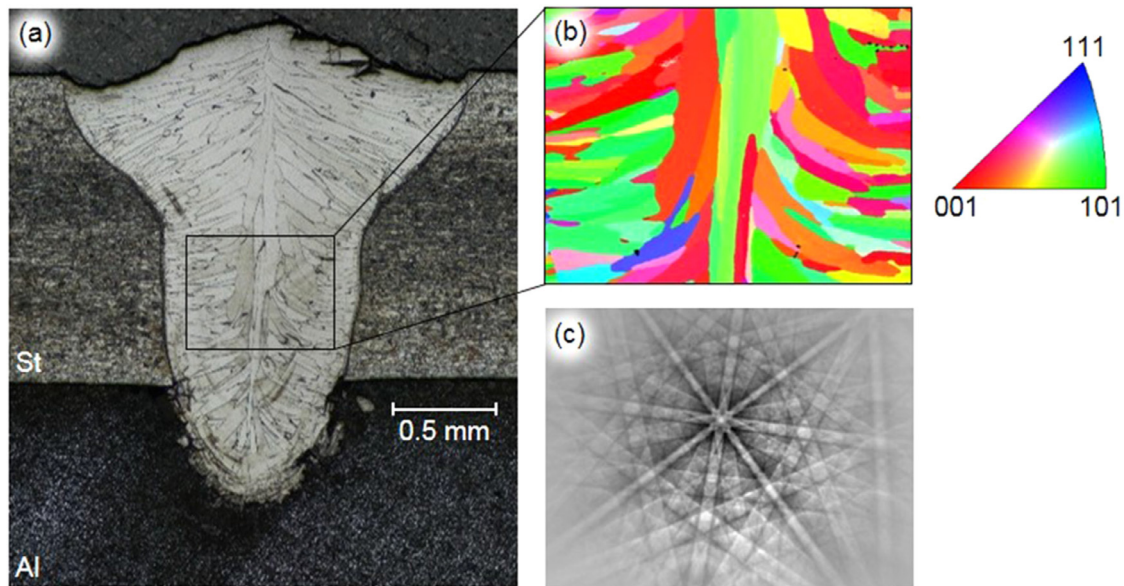


Fig. 22. EBSD of St-Al weld with penetration depths of $\sim 500 \mu\text{m}$. SEM of St/Al weld (a), inverse pole figure color map (b), Kikuchi diffraction pattern (c).

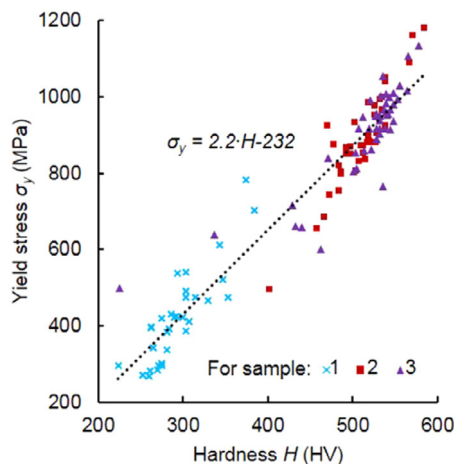


Fig. 23. Yield strength vs. hardness of St-Al weld metal. Samples 1-3 according to Fig. 7.

$\text{Fe}_{55}(\text{CrNiAl})_{45}$ and $\text{Fe}_{65}(\text{CrNiAl})_{35}$ also exhibit a single bcc structure, despite their VEC values of 7.25 and 7.42 respectively [41]. Stepanov et al. [31] found bcc structure in $\text{Fe}_{36}\text{Mn}_{21}\text{Cr}_{18}\text{Ni}_{14}\text{Al}_{10}$ alloy, which VEC is 7.19. Ma et al. [44] found a transition from bcc-fcc to bcc structure at $\text{VEC} < 7.35$. Thus, the phases cannot be precisely predicted using the existing empirical VEC rules. The VEC values corresponding to the transition between fcc, fcc-bcc mixture, and bcc should be further calibrated for the non-equiatom medium-entropy Fe-Cr-Ni-Mn-Al alloys. Calculation of the strengthening effect caused by the precipitates requires the knowledge of average particle size and their volume fraction [44]. However, the detailed formation of Ni-Al-rich precipitates within the St-Al weld metal is unclear and needs further investigation. Thus, modelling of the weld metal yield strength is up to now limited due to the uncertainty in the prediction of a crystalline structure as well as due to the lack of investigation on the precipitation kinetics. Therefore, the experimental characterization of the St-Al weld metal is inevitable.

The proposed in this study method for determination of the weld metal yield strength requires nano-indentation measurements and subsequent reverse analysis of the recorded load-depth curves. From the

engineering point of view, a fast estimation is of major importance. The yield strength can usually be correlated to the hardness, which can be obtained by conventional hardness tests. Fig. 23 plots yield strength versus hardness for the St-Al weld metal. The linear correlation is observed. The same correlation is inherent for many types of steel [54,55]. Thus, considering that Young's modulus and strain hardening exponent are nearly insensitive to Al concentration (Fig. 8), the elastic-plastic behavior of the St-Al welds with various penetration depths can be approximated based on the hardness measurements.

6.2. Influence of the St-Al weld material model

The influence of the weld formation on the overall structural behavior of the St-Al overlap joints is one of the central questions of the study. To demonstrate the importance of the developed weld material model WMM, the computed distortions were compared to the ones obtained by using the simplified weld material models. The first simplified model SWMM_1 assumes that the elastic-plastic behavior of the St-Al weld metal can be approximated by the mechanical properties of the steel. The second simplified model SWMM_2 uses the linear mixture rule to calculate the resulting mechanical properties.

The developed weld material model does not improve the computation accuracy concerning the transverse and longitudinal shrinkage. All obtained results are within the acceptable discrepancies from the experiments. However, the influence of the weld material model on the angular distortions is crucial. Fig. 24 plots the computed distortions as a function of welding speed. The graph also contains the experimentally measured distortions. The welding speed of 5.4 m/min results in solid-liquid bonding between the sheets without mixing inside the weld pool. In this case the weld does not affect the structural behavior of the joint and no difference in computed distortions is observed. With decreasing welding speed and hence increasing input energy, the mixing inside the weld pool occurs. It causes the formation of the St-Al weld metal and therefore deviations in the computed distortions using different weld material models. As can be seen from Fig. 24, the angular distortions are well captured by the simulations using the newly developed weld material model WMM. The computations with simplified weld material models completely failed to provide reliable results.

Despite the good accuracy achieved in the current work, more studies regarding the influence of Al concentration on the thermal expansion coefficient of the St-Al weld metal would be worthwhile. Further

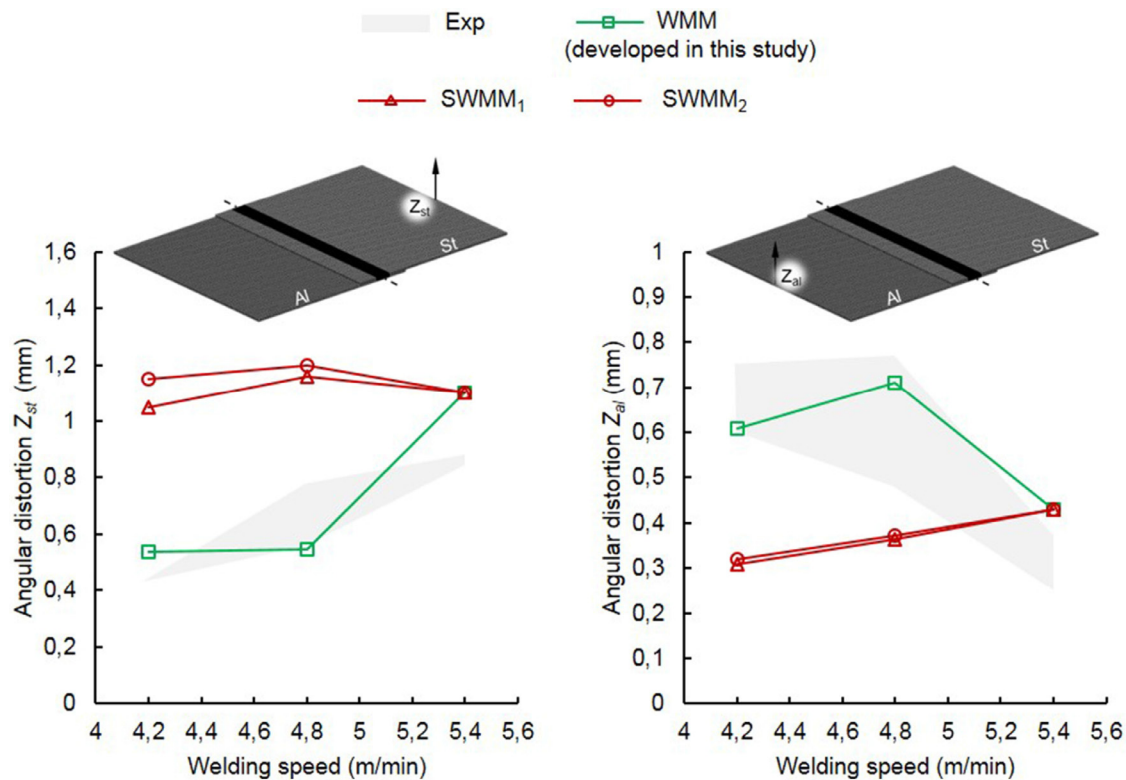


Fig. 24. Variation in prediction accuracy, when computing angular distortions using different weld material models. SWMM₁ approximates the elastic-plastic behavior of the St-Al weld metal by the mechanical properties of the steel, SWMM₂ uses the linear mixture rule to calculate the resulting mechanical properties and WMM is a developed weld material model as described in Section 3.2. $P_L = 3.75$ kW.

research could also focus on developing multi-physical models, similar to the one proposed by Wu et al. [56], for the exact calculation of the chemical compositions within the St-Al welds.

7. Conclusions

The study presented a numerical model for thermomechanical simulation of laser-welded dissimilar steel-aluminum (austenitic stainless steel 304 – 6082-T6 aluminum alloy) overlap joints. Indentation technique coupled with EDS analysis was used to characterize the elastic-plastic behavior of the St-Al weld metal as a function of the aluminum concentration. Using the obtained correlation, the weld material model was designed to consider the influence of the weld formation on the structural behavior of the St-Al overlap joints.

The following conclusions can be drawn from the work:

- The St-Al weld metal exhibits a gradual increase in hardness and yield strength with a rise in Al concentration from 0 to 9 at%. Further increase in Al concentration does not affect the strength significantly.
- The addition of Al in 304 stainless steel in the weld pool causes the transformation of the initial fcc structure into bcc and the formation of B2 Ni-Al nanoprecipitates. Since these precipitates are coherent with the bcc phase, the increase in bcc fraction results in substantial strengthening of the weld metal. Thus, it is suggested that the St-Al weld metal with Al concentration up to 9 at% exhibits a mixture of fcc and bcc phases with hardness and yield strength increase proportional to the fraction of the bcc phase. The strength saturation at Al concentrations beyond 9 at% is related to the formation of the fully bcc matrix. The bcc structure of the weld metal with Al concentration of 11 at% was experimentally confirmed through the EBSD analysis.

- There is a significant difference in the Al concentration in the upper and lower zones of the St-Al welds. Since the thermal expansion coefficient and elastic-plastic properties of the weld metal are functions of the Al concentration, the both zones should be considered in the computational model separately. The designed weld material model calculates the Al concentration in each zone using the empirical relations between the dilution of the alloys in the weld pool and aluminum distribution within the weld. Having the aluminum concentrations as inputs, the required sets of the elastic-plastic properties and thermal expansion coefficients are calculated for both zones. While heating, the structural behavior is governed by the properties of steel and aluminum, whereas during cooling the material response within the fusion zone is defined by the weld material model.
- Neither the approximation of the weld metal mechanical properties as the ones of the austenitic stainless steel nor the calculation based on the linear mixture rule allows computing the angular distortions of the St-Al overlap joints. A proper formulation of the weld material model, like the one presented in this study, is required.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Anton Evdokimov: Conceptualization, Methodology, Software, Investigation, Writing - original draft. **Nikolay Doynov:** Methodology, Software. **Ralf Ossenbrink:** Conceptualization, Writing - review & editing, Project administration, Funding acquisition. **Aleksei Obrosof:** Methodology, Investigation. **Sabine Weiß:** Writing - review & editing.

Vesselin Michailov: Writing - review & editing, Supervision, Funding acquisition.

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